Clemens Thalhammer

PhD Student · Applied Mathematics

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Education

ETH Zurich *Zurich, Switzerland*
PhD Applied Mathematics *since 2025*
- Supervisor: Habib Ammari

ETH Zurich *Zurich, Switzerland*
MSc Applied Mathematics *2023 - 2025*
- Focus in Mathematical Physics, Graduated with Distinction. Grade 5.78/6

ETH Zurich *Zurich, Switzerland*
BSc Mathematics *2020 - 2023*
- Grade 5.24/6

Research Experience

Imperial College London - Dept. of Mathematics *London, United Kingdom*
Advisors: Prof. Habib Ammari, Prof. Richard Craster, Dr. Bryn Davies *Summer 2024*
- MSc Thesis: "Computational Models for Quasiperiodic Differential Equations"

Teaching Experience

2023 **Numerical Methods**, Exam Preparation Course
2023 **Numerical Methods**, Teaching Assistant
2022 **Numerical Methods**, Teaching Assistant

Awards, Fellowships, & Grants

2025 **ETH Medal for an outstanding Master Thesis**, ETH Zurich

Presentations

Invited Talks

Summer 2026. **A Real-Space Formulation of the Zak Phase via Weyl m-Functions**. Invited talk: MetaMAT, Ajaccio, France.

Spring 2025. **Computational Models for Quasiperiodic ODEs**. Invited talk: Workshop on Metamaterials with Extra Dimensions, London, United Kingdom.

Leadership & Service

Seminar of Applied Mathemathics, ETH Zurich *Zurich, Switzerland*
Vice Group Organiser *since 2026*
Oversaw the hiring of teaching assistants and coordinated the planning and distribution of teaching duties among PhD students. Contributed to administrative processes related to course organisation and delivery, ensuring smooth coordination and execution.

VMM (Mid-level Faculty Association), ETH Zurich *Zurich, Switzerland*
Co-President *since 2025*
Represented early-career researchers in faculty governance and contributed to policy development and strategic decisions. Acted as a liaison between faculty leadership and members, ensuring transparent communication and engagement.

Peer Review

Journal of Differential Equations

Publications

Habib Ammari, Silvio Barandun, Yannick De Bruijn, Ping Liu, Clemens Thalhammer. 2025. Spectra and pseudo-spectra of tridiagonal k-Toeplitz matrices and the topological origin of the non-Hermitian skin effect. Journal of Physics A: Mathematical and Theoretical 58(20):205201.

Bryn Davies, Clemens Thalhammer. 2025. Convergence of supercell and superspace methods for computing spectra of quasiperiodic operators. Proceedings of the Royal Society A 481(2319):20240894.

Habib Ammari, Silvio Barandun, Clemens Thalhammer, Alexander Uhlmann. 2026. Competing Edge and Bulk Localization in Nonreciprocal Disordered Systems. Multiscale Modeling & Simulation 24(2):456–473.

Habib Ammari, Clemens Thalhammer. 2026. A real-space formulation of the Zak phase via Weyl m-functions. Journal of Physics A: Mathematical and Theoretical.

Preprints

Habib Ammari, Clemens Thalhammer, Alexander Uhlmann. 2025. Uniform Hyperbolicity, Bandgaps and Edge Modes in Aperiodic Systems of Subwavelength Resonators. arXiv:2509.22417.

Habib Ammari, Clemens Thalhammer. 2026. Perturbative Approach to Nonlinear Capacitance Matrix Formulations. arXiv:2606.20821.

Elena Cherkaev, Bryn Davies, Sébastien Guenneau, Clemens Thalhammer, Niklas Wellander. 2026 High-frequency spectral asymptotics and homogenization for quasiperiodic operators.arXiv:2607.03064